

in CW direction. This cycle repeats until S3 or S4 is pressed. You can control the head movement anytime during run time using S3 or S4.

Left and right movements. M1, S1 and S3 are responsible for left and right movements of the spray head. RL1 is energised for CCW direction and RL2 for CW direction. Refer Fig. 5 for different movements of the spray head.

Down and up movements. M2, S2 and S4 are responsible for down and up movements. RL3 and RL4 are energised for down and up directions, respectively.

Relay connections. The 3-phase induction motor connections in CW

and CCW directions are shown in Fig. 6. RL1 and RL2 are used to drive M1 in CW and CCW directions, respectively, via contactor1. It has two separate individual contactors, CONT1 (A) and CONT1 (B).

RL3 and RL4 are used to drive M2 via contactor2. It also has two separate individual contactors, CONT2(A) and CONT2(B).

RL5 is used to drive the nozzle. Its circuit is shown in Fig. 7. An optional relay (RL6), as shown in the circuit, is not used here. S5 is used in auto mode to scan and spray the whole surface of the sheet a number of times, until the MCU is forced to reset, or the circuit is switched off.

Software.

Software is written in C program. Kiel µVision is used to generate hex code. Before using AT89C51 MCU in the assembled PCB, burn the hex code into the MCU using a suitable programmer.

Construction and testing

An actual-size PCB layout of the thin-film deposition technique using MCU is shown in Fig. 8 and its components layout in Fig. 9. After assembling the circuit on the PCB, connect the relays to the respective motor and contactor according to the circuit given in Fig. 6.

Also, connect the nozzle to RL5 according to the circuit shown in Fig. 7. Use external wires to con-

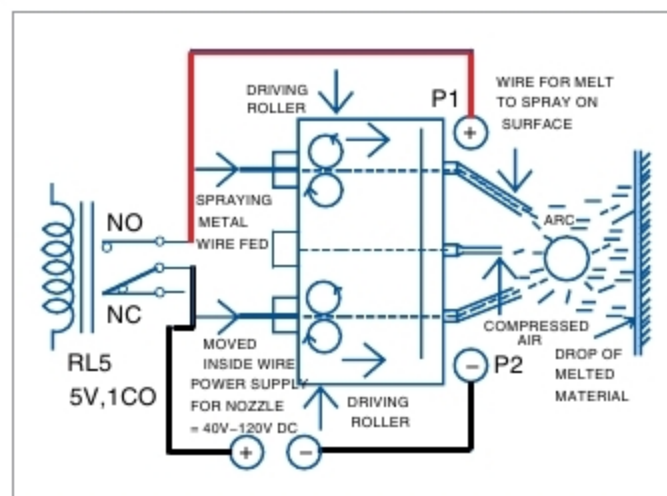


Fig. 7: Nozzle connection with RL5

PARTS LIST

Semiconductors:

IC1	- AT89C51 MCU
D1-D6	- 1N4007 rectifier diode
T1-T6	- BC327 pnp transistor
LED1-LED6	- 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

R1-R6	- 1-kilo-ohm
R7-R10, R12	- 10-kilo-ohm
R11	- 8.2-kilo-ohm
R13	- 470-ohm
VR1	- 10-kilo-ohm preset

Capacitors:

C1	- 1µF, 16V electrolytic
C2, C3	- 22pF, ceramic disk

Miscellaneous:

CON1	- 2-pin terminal connector
S1-S4	- Limit switch
S5, S6	- Tactile switch
RL1-RL6	- 5V, single changeover relay
LCD1	- 16x2 LCD
XTAL1	- 11.0592MHz crystal oscillator
M1, M2	- 10rpm 3-phase motor
CONT1	- 3-phase, 2-coil 110V, 240V, 415V AC contactor
CONT2	- 3-phase, 2-coil 110V, 240V, 415V AC contactor
	- Spray painting setup
	- Jumper wires and cables
	- Nozzle with 40V-120V DC supply
	- 5V regulated power supply
	- 3-phase supply

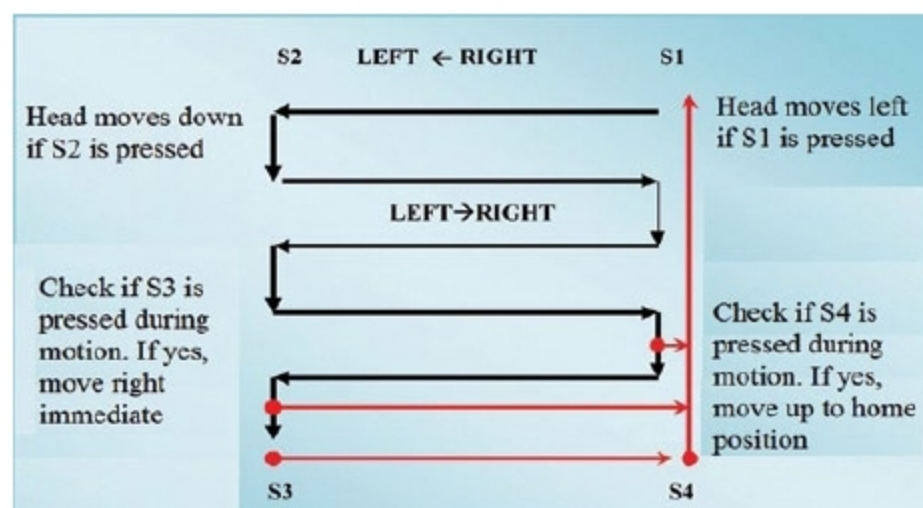


Fig. 5: Movement of spray head in x and y directions

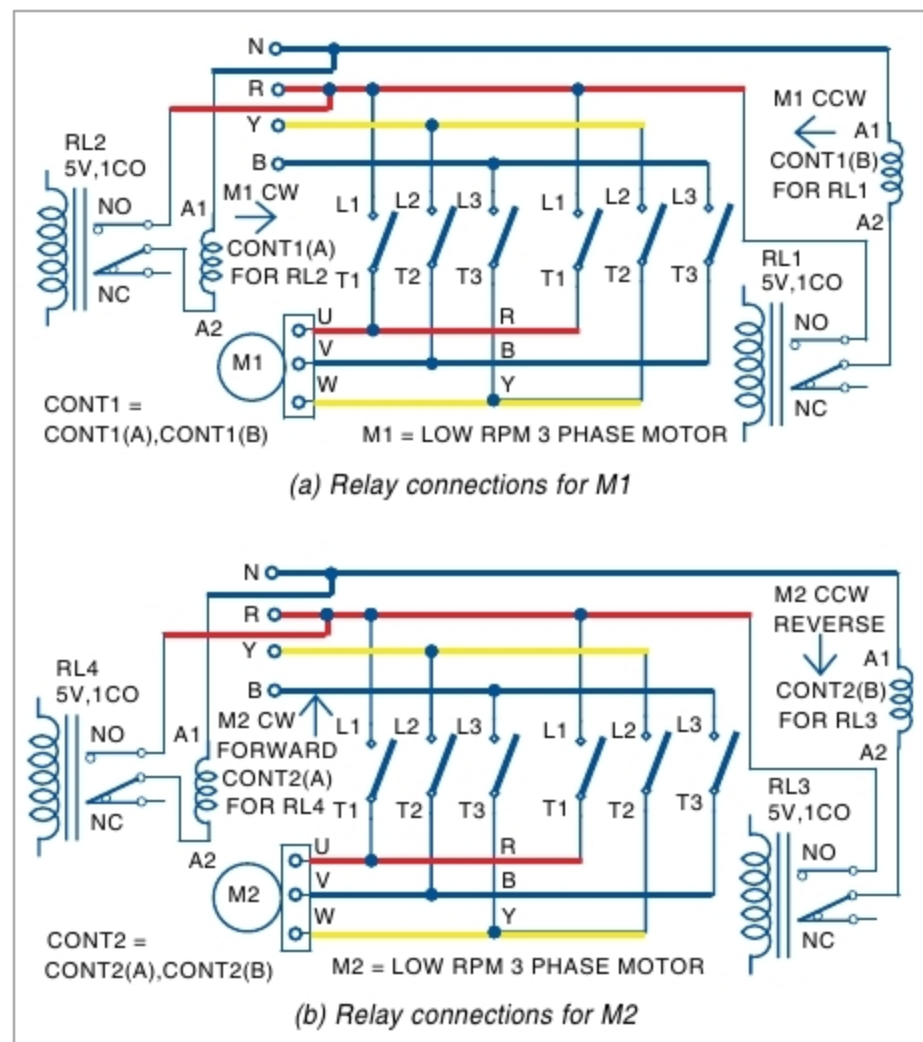


Fig. 6: Motor connections with contactors and relays

nect 3-phase motors, contactors, nozzle and related circuitry. Mount limit switches at appropriate locations so that S1 is on the right and S2 is on the left, as shown in the setup in Fig. 1.

Testing procedure

1. Switch on the power supply to the circuit. The head should be at the top-right side near S1.
2. Configure the vertical down movement (y direction) of M2 after completion of movement in both left